Slides and example codes are available:
https://github.com/mint-lab/cv_tutorial

Image Geometry

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Review) Absolute Camera Pose Estimation

- Example) **Pose estimation (book) + camera calibration – initially given** K `[pose_estimation_book3.py]`

(due to wrong initial K)

- Load the cover image
- Extract ORB features
- Build **3D object points** $\mathbf{X}_1, \mathbf{X}_2, \ldots$

Pass tentative matches

- Grab an input image
- Extract **ORB features** $\mathbf{x}_1, \mathbf{x}_2, \ldots$
- Match them to the object features

[Outlier Rejection]
- Remove wrong matches using `cv::findHomography()`

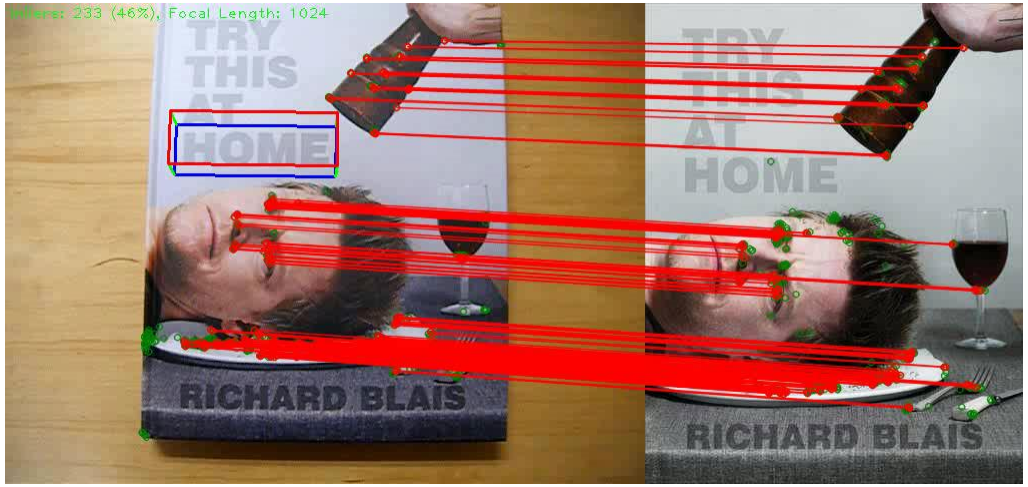
(free from K)

Pass correct matches

[Camera Pose Estimation]
- Estimate camera parameters using `cv::calibrateCamera()`

[Camera Pose Update]
- Intrinsic: K (initially known)
- Extrinsic: R, $\mathbf{t}$

Draw the box using `cv::projectPoints()`



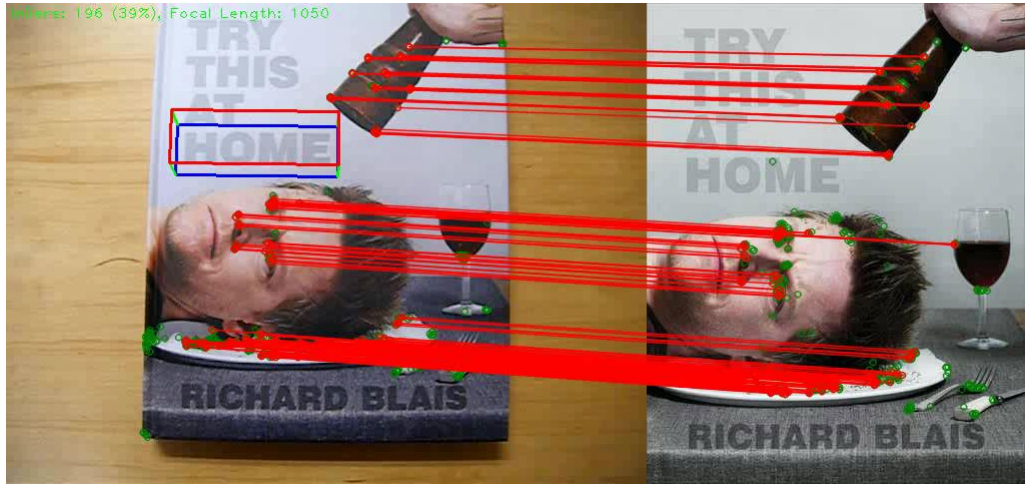
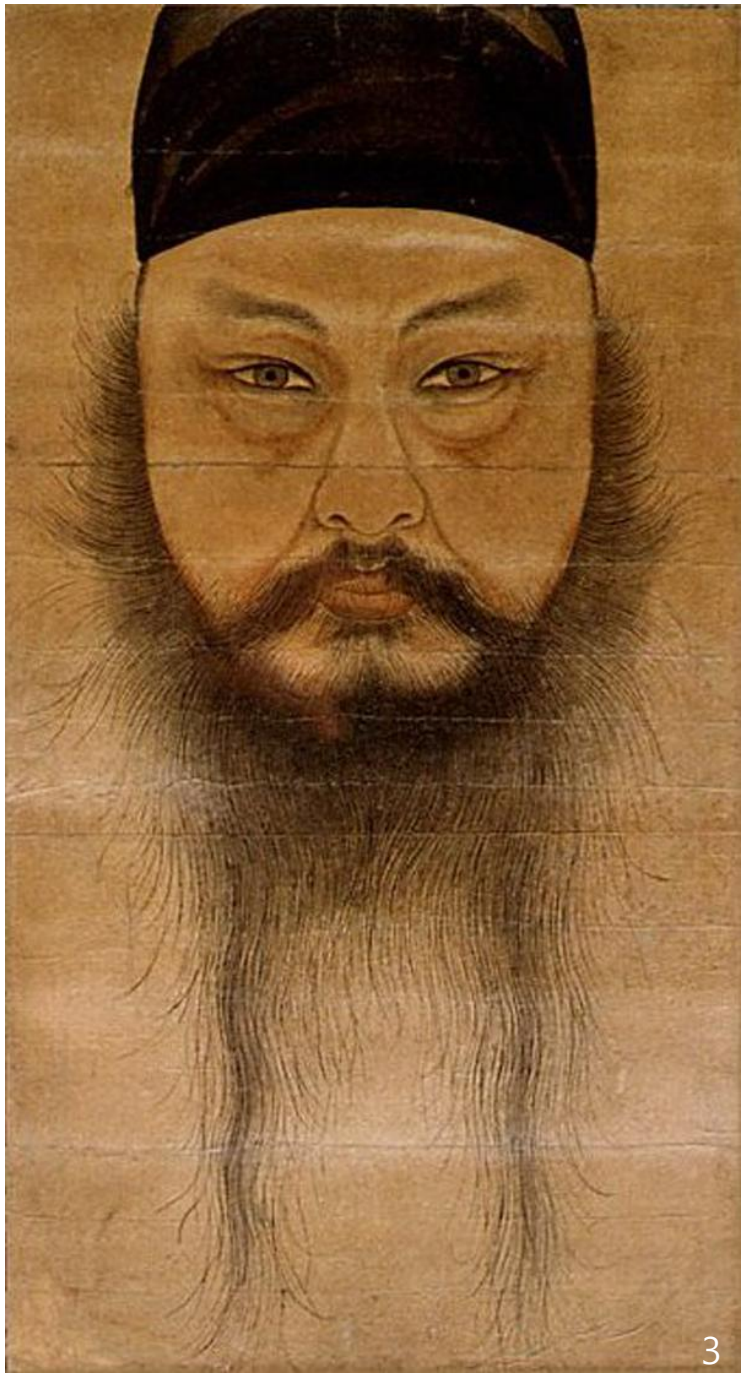


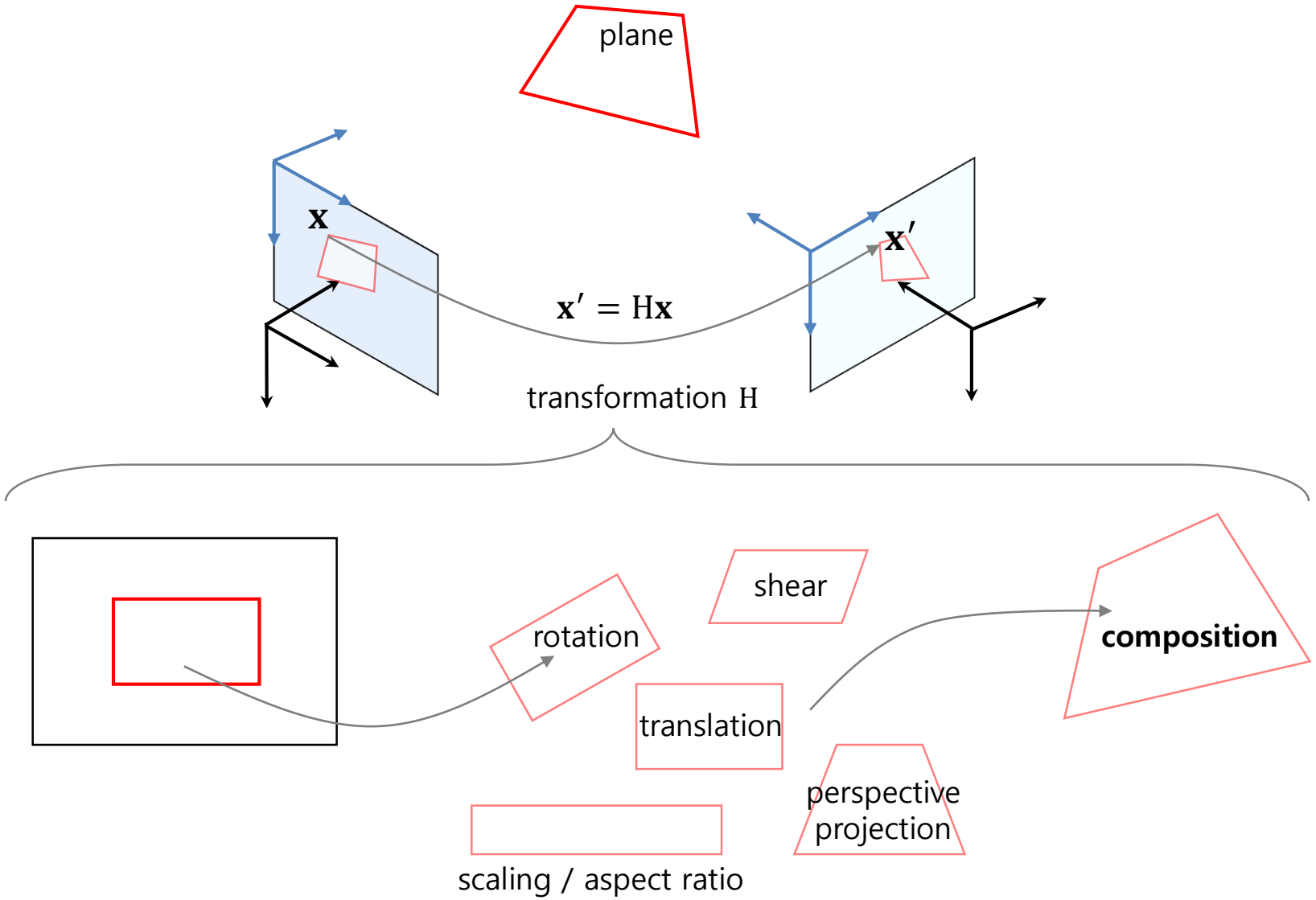
Table of Contents: Two-view Geometry

- **Planar Homography**
- ~~**Epipolar Geometry**~~
  - ~~Epipolar constraint~~
  - ~~Fundamental and essential matrix~~
- ~~**Relative Camera Pose Estimation**~~
- **Triangulation**

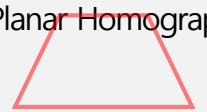


윤두서(1668-1715) 자화상, 국보 제240호
Korean National Treasure No. 240

Planar Homography



Planar Homography

	Euclidean Transformation (a.k.a. Rigid Transform)	**Similarity Transformation**	**Affine Transformation**	**Projective Transformation** (a.k.a. Planar Homography) 
Matrix Forms H	$\begin{bmatrix} \cos\theta & -\sin\theta & t_x \\ \sin\theta & \cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ v_1 & v_2 & 1 \end{bmatrix}$
DOF	3	4	6	8
Transformations				
- rotation	O	O	O	O
- translation	O	O	O	O
- scaling	X	O	O	O
- aspect ratio	X	X	O	O
- shear	X	X	O	O
- perspective projection	X	X	X	O
Invariants				
- length	O	X	X	X
- angle	O	O	X	X
- ratio of lengths	O	O	X	X
- parallelism	O	O	O	X
- incidence	O	O	O	O
- cross ratio	O	O	O	O

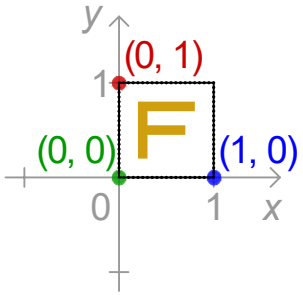
OpenCV APIs			`cv.getAffineTransform()`
			`cv.estimateRigidTransform()`
			-
			`cv.warpAffine()`
				`cv.getPerspectiveTransform()`
				-
				`cv.findHomography()`
				`cv.warpPerspective()`

Note) Similarly **3D transformations** (3D-3D geometry) are represented as **4x4 matrices**.

Note) Affine Transformation

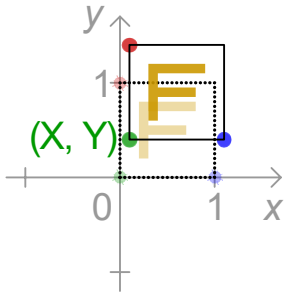
No change

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



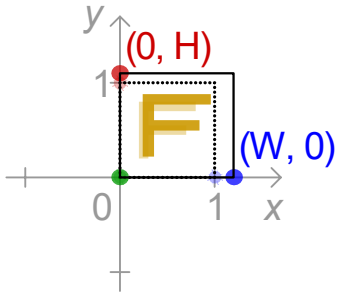
Translate

$$\begin{bmatrix} 1 & 0 & X \\ 0 & 1 & Y \\ 0 & 0 & 1 \end{bmatrix}$$



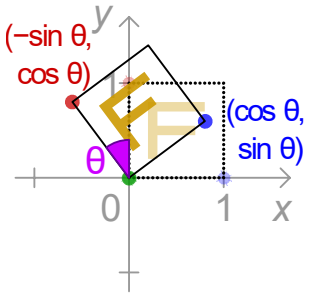
Scale about origin

$$\begin{bmatrix} W & 0 & 0 \\ 0 & H & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



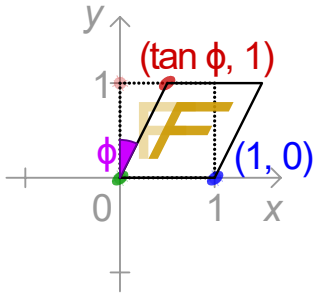
Rotate about origin

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



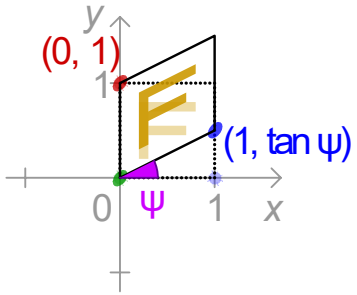
Shear in x direction

$$\begin{bmatrix} 1 & \tan\phi & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



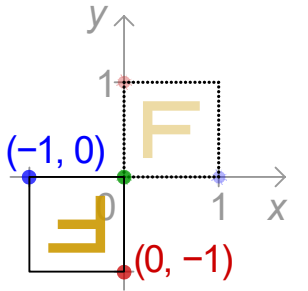
Shear in y direction

$$\begin{bmatrix} 1 & 0 & 0 \\ \tan\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



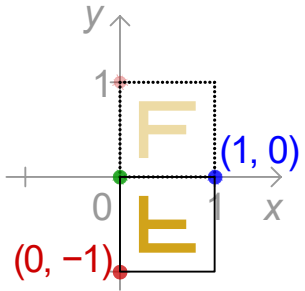
Reflect about origin

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Reflect about x-axis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Reflect about y-axis

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

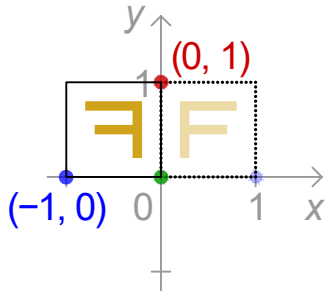
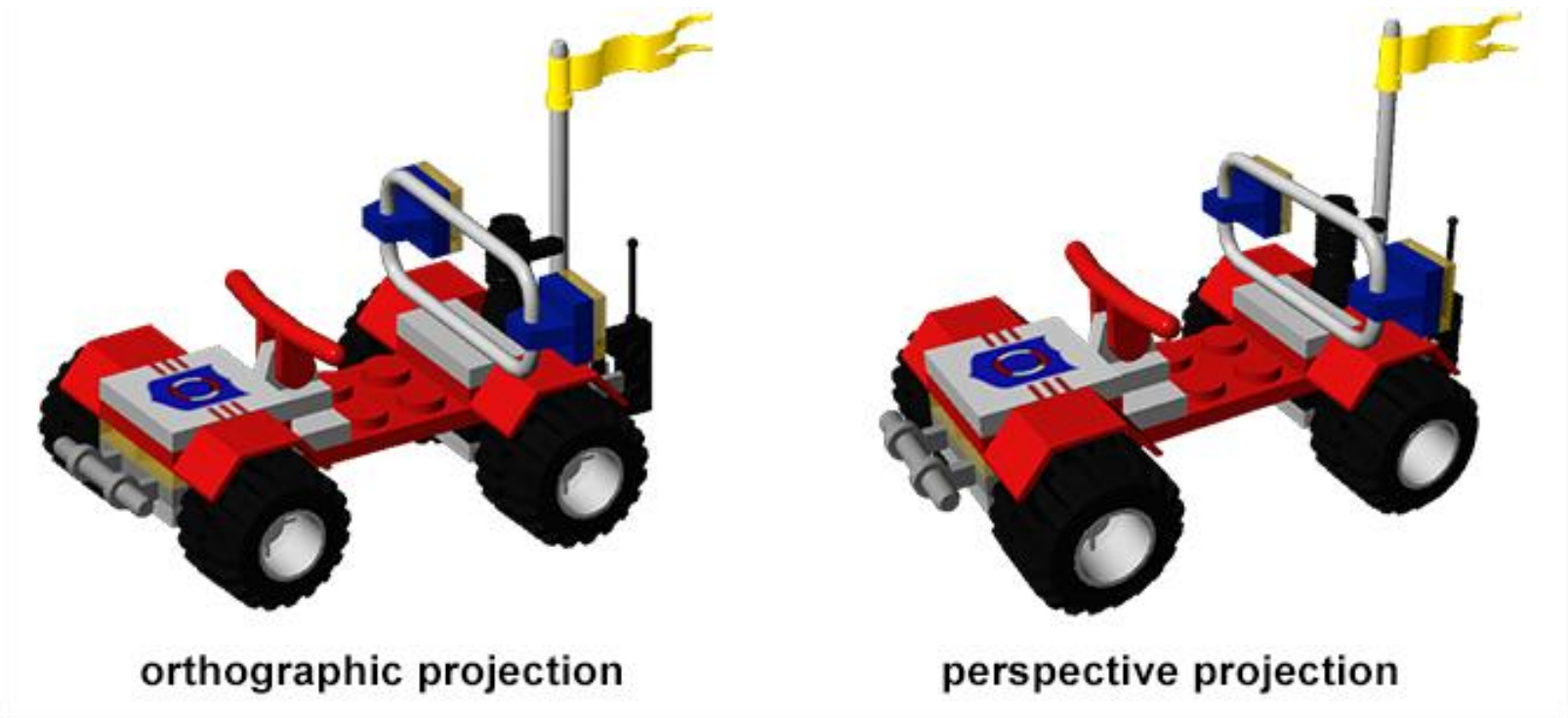


Image: Wikipedia

Note) Affine Transformation



- **Q) What is the difference between two images?**
  - **Parallel projection** vs. Perspective projection
    - Affine camera (a.k.a. weak perspective camera; $f = \infty$)
    - Less natural, but (sometimes) useful for technical visualization (e.g. CAD)

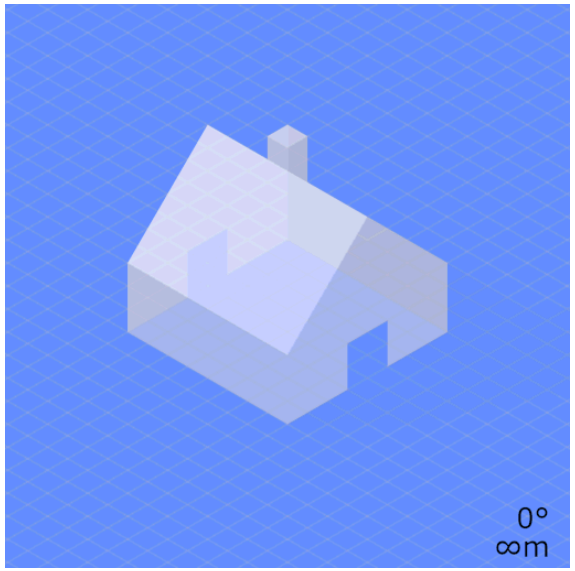
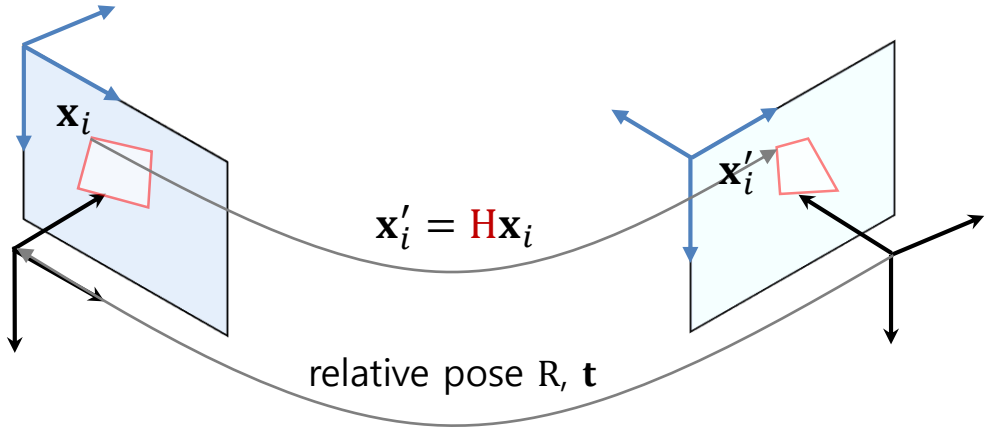


Image: Scatchpixel, Wikipedia

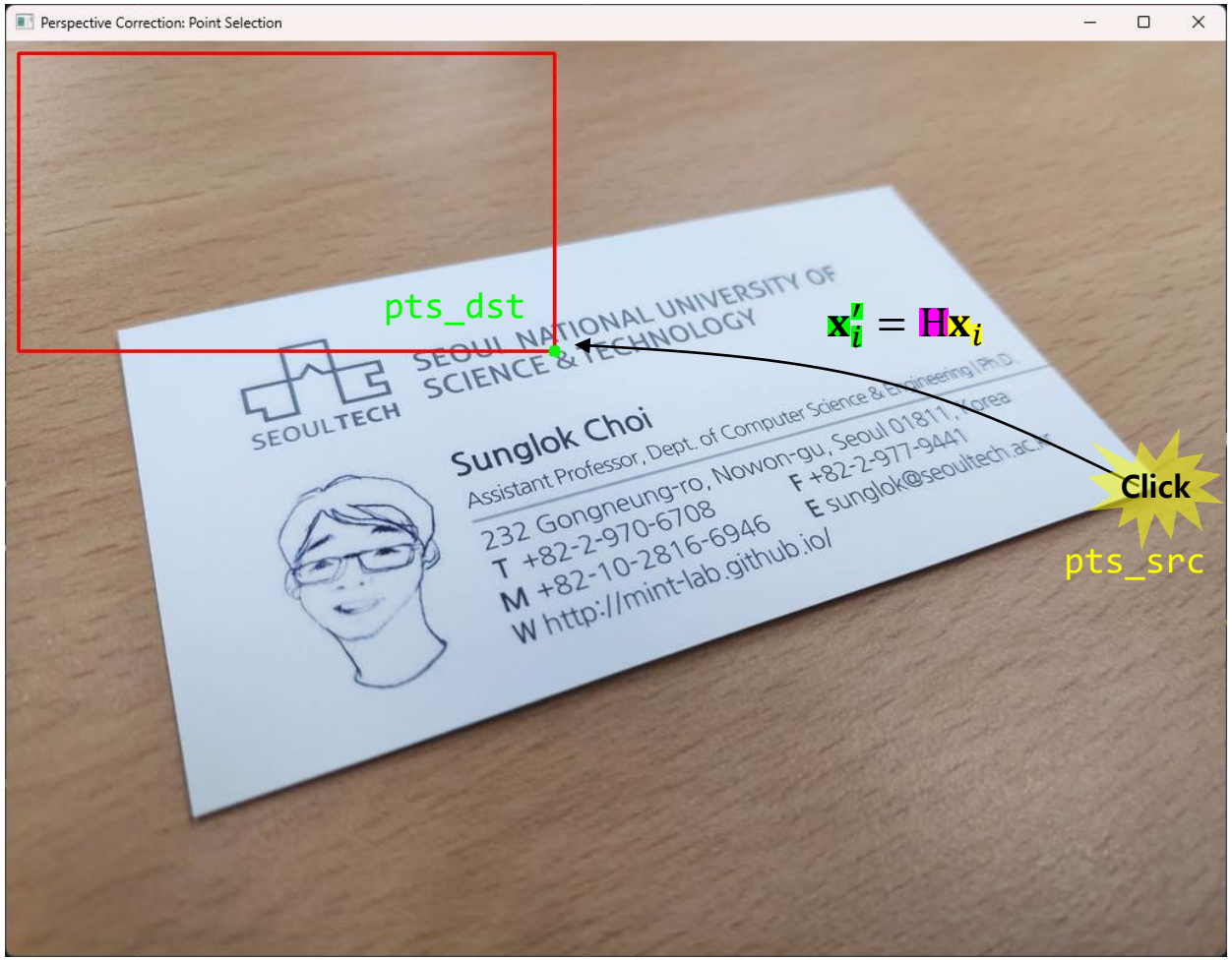
Planar Homography

- **Planar homography estimation**
  - Unknown: Planar homography H (8 DOF)
  - Given: Point correspondence $(\mathbf{x}_1, \mathbf{x}'_1)$, …, $(\mathbf{x}_n, \mathbf{x}'_n)$
  - Constraints: n x projective transformation $\mathbf{x}'_i = \mathrm{H}\mathbf{x}_i$
  - Solutions ($n \geq 4$) → 4-point algorithm
    - OpenCV: `cv.getPerspectiveTransform()` and `cv.findHomography()`
    - Note) Simpler transformations requires fewer correspondences.
      - Affine ($n \geq 3$), similarity ($n \geq 2$), Euclidean ($n \geq 2$)
  - Note) Planar homography can be decomposed as relative camera pose.
    - OpenCV: `cv.decomposeHomographyMat()`
    - The decomposition needs to know camera matrices.



Planar Homography

- Example) **Perspective distortion correction** `[perspective_correction.py]`



Planar Homography

- Example) **Perspective distortion correction** [perspective_correction.py]

```python
def mouse_event_handler(event, x, y, flags, param):
    if event == cv.EVENT_LBUTTONDOWN:
        param.append((x, y))

if __name__ == '__main__':
    img_file = '../data/sunglok_card.jpg'
    card_size = (450, 250)
    offset = 10

    # Prepare the rectified points
    pts_dst = np.array([[0, 0], [card_size[0], 0], [0, card_size[1]], [card_size[0], card_size[1]]])

    # Load an image
    img = cv.imread(img_file)

    # Get the matched points from mouse clicks
    pts_src = []
    wnd_name = 'Perspective Correction: Point Selection'
    cv.namedWindow(wnd_name)
    cv.setMouseCallback(wnd_name, mouse_event_handler, pts_src)
    while len(pts_src) < 4:
        img_display = img.copy()
        cv.rectangle(img_display, (offset, offset), (offset + card_size[0], offset + card_size[1]), (0, 0, 255), 2)
        idx = min(len(pts_src), len(pts_dst))
        cv.circle(img_display, offset + pts_dst[idx], 5, (0, 255, 0), -1)
        cv.imshow(wnd_name, img_display)
```

Planar Homography

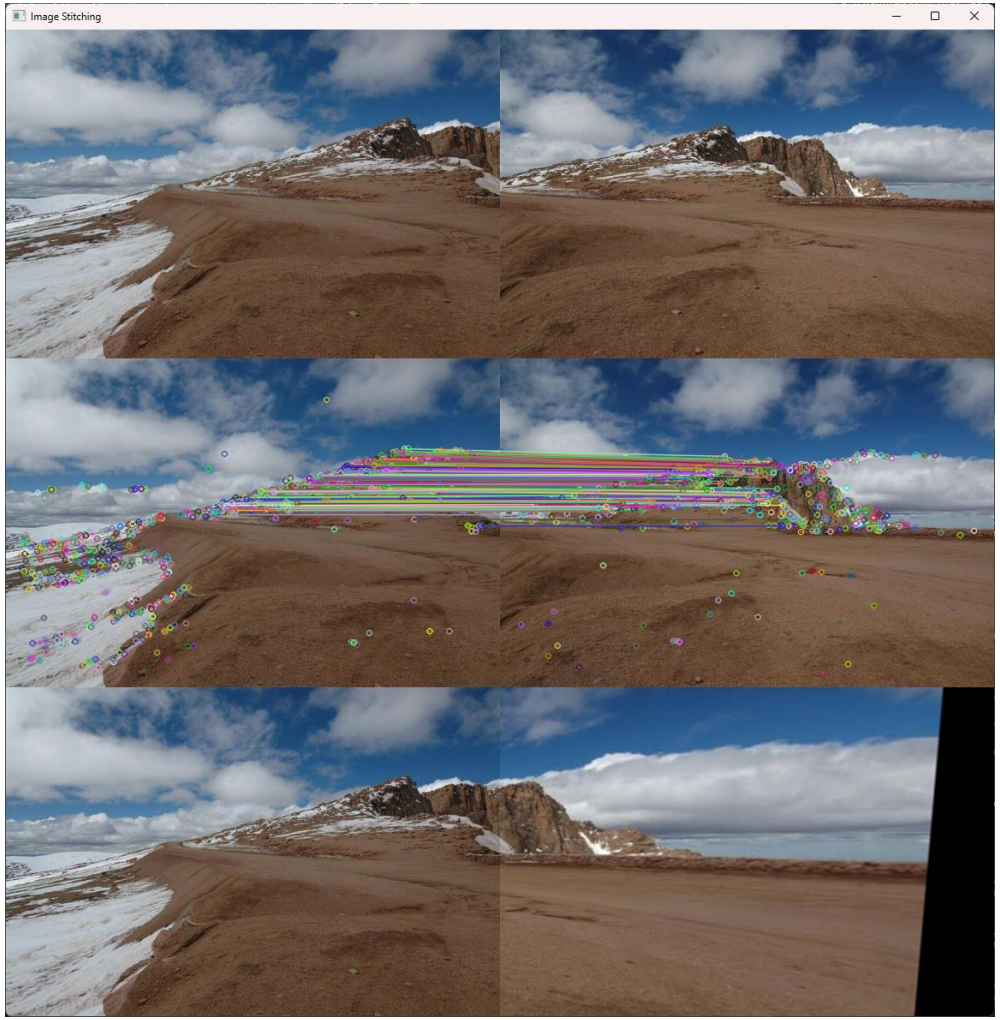
- Example) **Perspective distortion correction** [perspective_correction.py]

```python
if __name__ == '__main__':
    img_file = '../data/sunglok_card.jpg'
    card_size = (450, 250)
    offset = 10

    # Prepare the rectified points
    pts_dst = np.array([[0, 0], [card_size[0], 0], [0, card_size[1]], [card_size[0], card_size[1]]])

    # Load an image
    img = cv.imread(img_file)

    # Get the matched points from mouse clicks
    pts_src = []
    ...

    if len(pts_src) == 4:
        # Calculate planar homography and rectify perspective distortion
        H, _ = cv.findHomography(np.array(pts_src), pts_dst)
        img_rectify = cv.warpPerspective(img, H, card_size)

        # Show the rectified image
        cv.imshow('Perspective Correction: Rectified Image', img_rectify)
        cv.waitKey(0)

    cv.destroyAllWindows()
```

Planar Homography

- Example) **Planar image stitching** [image_stitching.py]

```
# Load two images
img1 = cv.imread('../data/hill01.jpg')
img2 = cv.imread('../data/hill02.jpg')

# Retrieve matching points
fdetector = cv.BRISK_create()
keypoints1, descriptors1 = fdetector.detectAndCompute(img1, None)
keypoints2, descriptors2 = fdetector.detectAndCompute(img2, None)

fmatcher = cv.DescriptorMatcher_create('BruteForce-Hamming')
match = fmatcher.match(descriptors1, descriptors2)

# Calculate planar homography and merge two images
pts1, pts2 = [], []
for i in range(len(match)):
    pts1.append(keypoints1[match[i].queryIdx].pt)
    pts2.append(keypoints2[match[i].trainIdx].pt)
pts1 = np.array(pts1, dtype=np.float32)
pts2 = np.array(pts2, dtype=np.float32)

H, inlier_mask = cv.findHomography(pts2, pts1, cv.RANSAC)
img_merged = cv.warpPerspective(img2, H, (img1.shape[1]*2, img1.shape[0]))
img_merged[:,:img1.shape[1]] = img1 # Copy

# Show the merged image
img_matched = cv.drawMatches(img1, keypoints1, img2, keypoints2, match, None, None, None,
```



Planar Homography

- Example) **Planar image stitching → *Gigapixel* images** (with 10^9+ pixels)

Image: [Wikimedia Commons]

Planar Homography

- Example) ~~Planar~~ **image stitching** → ***Panoramic*** **images** (with wide-angle view)

Image: OpenPano

Image: Seada Haskić, John David Lynn Brosz, Wikipedia

Beyond Planar Homography

- **Projection models for *panoramic* images** (with wide-angle view)

3D Model

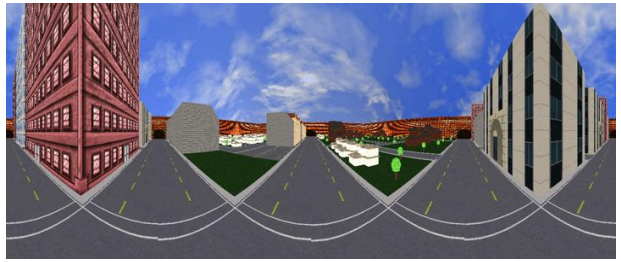
Planar/Perspective view

$z = \text{const}$

+ Natural (preserve straight lines)
- Limited field-of-view (max <180°)

Cylindrical view

$x^2 + z^2 = \text{const}$

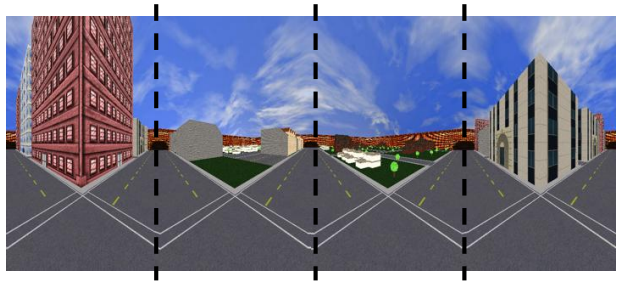


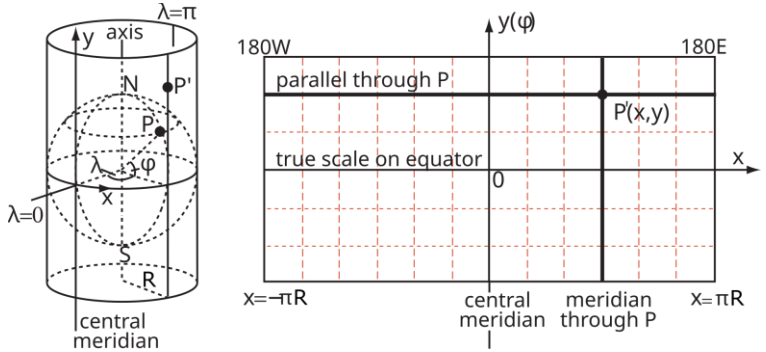
Spherical view

$x^2 + y^2 + z^2 = \text{const}$

Cubic view

$-\text{const} \le x \le +\text{const}$
$-\text{const} \le y \le +\text{const}$
$z = \pm\text{const}$ …





How do we represent a sphere in 2D?
e.g. Mercator projection

Planar Homography

- Example) **2D video stabilization** [video_stabilization.py]

```
# Open a video and get the reference image and feature points
video = cv.VideoCapture('../data/traffic.avi')

_, gray_ref = video.read()
if gray_ref.ndim >= 3:
    gray_ref = cv.cvtColor(gray_ref, cv.COLOR_BGR2GRAY)
pts_ref = cv.goodFeaturesToTrack(gray_ref, 2000, 0.01, 10)

# Run video stabilization
while True:
    # Read an image from `video`
    valid, img = video.read()
    if not valid:
        break
    if img.ndim >= 3:
        gray = cv.cvtColor(img, cv.COLOR_BGR2GRAY)
    else:
        gray = img.copy()

    # Extract optical flow and calculate planar homography
    pts, status, error = cv.calcOpticalFlowPyrLK(gray_ref, gray, pts_ref, None)
    H, inlier_mask = cv.findHomography(pts, pts_ref, cv.RANSAC)

    # Synthesize a stabilized image
    warp = cv.warpPerspective(img, H, (img.shape[1], img.shape[0]))
```

A shaky CCTV video

(Red: Inlier, Green: Outlier)

Planar Homography

- Assumption) **The same <u>plane</u> is observed from two viewpoints.**
  - Perspective distortion correction: A complete plane

Planar Homography

- Assumption) **A <u>plane</u> is observed by two views.**
  - Perspective distortion correction: A complete plane
  - Planar image stitching: An approximated plane (← distance ≫ depth variation)
  - 2D video stabilization: An approximated plane (← small motion)

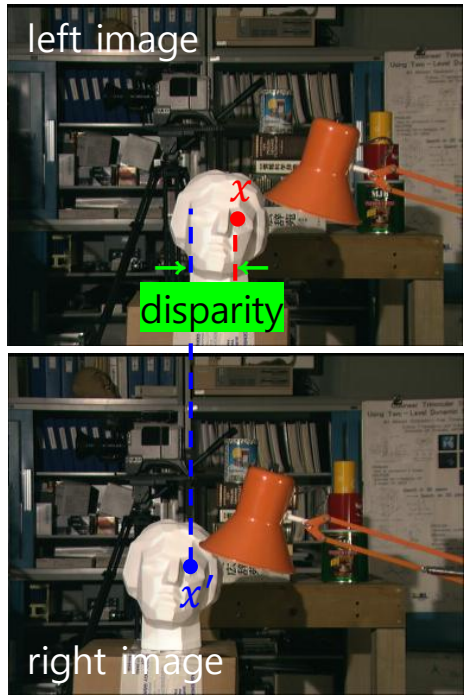
Triangulation

- **Triangulation** (point localization)
  - Unknown: Position of a 3D point $\mathbf{X}$ (3 DOF)
  - Given: Point correspondence $(\mathbf{x}, \mathbf{x}')$, camera matrices $(\mathrm{K}, \mathrm{K}')$, and relative pose $(\mathrm{R}, \mathbf{t})$
  - Constraints: $\mathbf{x} = \mathrm{K}\,[\,\mathrm{I} \mid \mathbf{0}\,]\,\mathbf{X} = \mathrm{P}\,\mathbf{X}$, $\mathbf{x}' = \mathrm{K}'\,[\,\mathrm{R} \mid \mathbf{t}\,]\,\mathbf{X} = \mathrm{P}'\,\mathbf{X}$
  - Solution
    - OpenCV `cv.triangulatePoints()`
  - Special case) Stereo cameras

$$\mathrm{R} = \mathrm{I}_{3\times3},\ \mathbf{t} = \begin{bmatrix} -b \\ 0 \\ 0 \end{bmatrix},\ \text{and } \mathrm{K} = \mathrm{K}' = \begin{bmatrix} f & 0 & c_x \\ 0 & f & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where b is the baseline (distance between two cameras)

$$\therefore Z = \frac{f}{x - x'} b,\ X = \frac{x - c_x}{f} Z,\ \text{and } Y = \frac{y - c_y}{f} Z$$



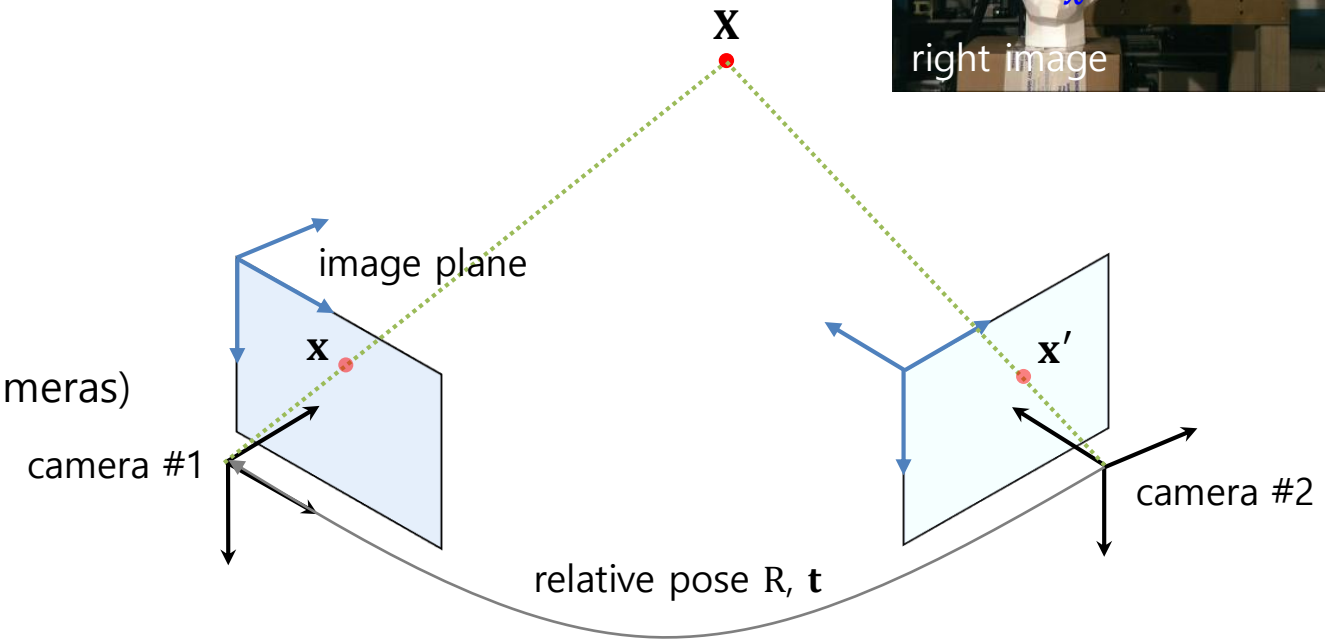


Image: Middlebury Stereo Vision Page

Triangulation

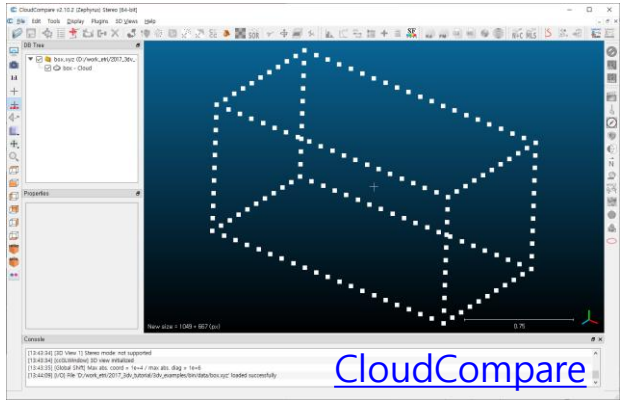
- Example) **Triangulation**

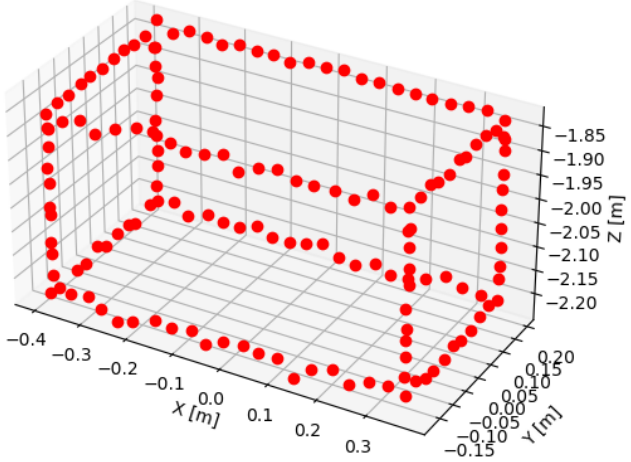
```
f, cx, cy = 1000., 320., 240.
pts0 = np.loadtxt('../data/image_formation0.xyz')[:,:2]
pts1 = np.loadtxt('../data/image_formation1.xyz')[:,:2]
output_file = '../data/triangulation.xyz'

# Estimate the relative pose of two view
F, _ = cv.findFundamentalMat(pts0, pts1, cv.FM_8POINT)
K = np.array([[f, 0, cx], [0, f, cy], [0, 0, 1]])
E = K.T @ F @ K
_, R, t, _ = cv.recoverPose(E, pts0, pts1)

# Reconstruct 3D points (triangulation)
P0 = K @ np.eye(3, 4, dtype=np.float32)
Rt = np.hstack((R, t))
P1 = K @ Rt
X = cv.triangulatePoints(P0, P1, pts0.T, pts1.T)
X /= X[3]
X = X.T

# Write the reconstructed 3D points
np.savetxt(output_file, X)
```

$$\mathbf{x} = \mathbf{K}\,[\,\mathbf{I}\,|\,\mathbf{0}\,]\,\mathbf{X} = \mathbf{P}\,\mathbf{X}$$

$$\mathbf{x}' = \mathbf{K}'\,[\,\mathbf{R}\,|\,\mathbf{t}\,]\,\mathbf{X} = \mathbf{P}'\,\mathbf{X}$$

A point cloud: `data/box.xyz`





`output_file`: `data/triangulation.xyz`

Summary

- **Planar Homography**: $\mathbf{x}_i' = \mathrm{H}\mathbf{x}_i$
  - Example) Perspective distortion correction
  - Example) Planar image stitching
  - Example) 2D video stabilization
- **Triangulation**: Finding $\mathbf{X}$ (3 DOF)
  - Example) Triangulation